

A Machine Learning Framework for Multi-Class Diabetes Prediction Using Stacking Ensemble and SHAP Explainability

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Abstract—Diabetes mellitus, one of the most prevalent chronic metabolic diseases globally as of 2022, affects approximately 830 million adults and its incidence is still increasing. Hence, early and accurate prediction is crucial to execute timely intervention, as timely action can save lives. Machine learning (ML) methods have shown great promise in predicting diabetes, yet most existing approaches are limited by binary classification on the Pima Indian Diabetes Dataset (768 samples). In this paper, we propose a comprehensive ML-based framework to classify multi-class diabetes types (No Diabetes, Prediabetes, and Diabetes) based on a large-scale CDC Behavioral Risk Factor Surveillance System (BRFSS) dataset of 458,560 records (after duplicates are removed from an original count of 641,109) with 41 features (21 original + 20 engineered). We systematically compare eight baseline ML algorithms: Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, XGBoost, and LightGBM, alongside CatBoost, a TabNet deep learning model, and a Multi-Layer Perceptron neural network as benchmarks. We present both a weighted AUC-based ensemble and a stacking ensemble with a logistic regression meta-learner, and employ SHAP (SHapley Additive exPlanations) for model interpretability. XGBoost, LightGBM, CatBoost, and TabNet are trained with GPU acceleration (NVIDIA RTX 3070) support. Our stacking ensemble outperforms the tuned individual models, achieving an accuracy of 95.24%, an F1-score of 0.9523, and an AUC-ROC of 0.9905. SHAP analysis shows that BMI-derived features, general health status, age–health interactions, and composite risk scores are the top predicting features across all classes, elucidating clinically actionable insights.

Index Terms—Diabetes Prediction, Machine Learning, Multi-Class Classification, Stacking Ensemble, CatBoost, TabNet, SHAP Explainability, XGBoost, LightGBM, CDC BRFSS, Feature Engineering, GPU Acceleration

I. INTRODUCTION

Diabetes mellitus is one of the most common challenges in the world of 21st-century global health. According to the World Health Organization (WHO), around 830 million adults had diabetes in 2022, a number that has more than quadrupled since 1990 [1]. According to statistics from the International Diabetic Federation (IDF), there are currently 589 million adults aged 20–79 with diabetes, with estimates of this number climbing to 853 million in 2050 [2]. Diabetes and diabetic nephropathy alone claimed the lives of over two million

people in 2021 [1]. Of particular concern to clinicians is that precarious stage before diabetes, called prediabetes. Blood glucose levels are high but do not yet cross the diagnostic threshold into full-blown diabetes. The early identification of prediabetes is a crucial opportunity for intervention through lifestyle changes and medical treatment.

Machine learning (ML) techniques have shown great promise lately in predicting diabetes risk from indicators of health; however, there is a careful examination of the literature today reveals three major troubles with this approach. First, most of the research is based on the Pima Indian Diabetes Dataset (PIDD), which consists of only 768 samples of a specific demographic (females from Pima Indian heritage, aged ≥ 21). This reduces both generalizability and clinical relevance to a significant extent [3]–[5]. Second, most existing work frames diabetes prediction as a binary (diabetic vs. non-diabetic) classification problem, completely overlooking the clinically important prediabetes stage, thereby failing to capture the progressive nature of the disease and missing the most actionable intervention point [6], [7]. Third, although high accuracy may be claimed by individual studies, comprehensive comparisons across diverse algorithms using consistent evaluation frameworks are rare, and clinical adoption of ML models requires interpretability, understanding *why* a model makes a specific prediction, which most studies do not address [6].

This paper addresses these gaps with the following contributions:

- A thorough evaluation of **eight baseline ML algorithms, CatBoost, TabNet, and MLP** on the CDC BRFSS dataset containing **458,560 records** (after deduplication) across three classes (No Diabetes, Prediabetes, Diabetes), making this one of the largest-scale multi-class diabetes prediction studies to date.
- We propose and compare two ensemble strategies: a **weighted AUC-based soft-voting ensemble** (extending the binary approach of [8] to multi-class) and a **stacking ensemble** with five GPU-accelerated base learners and a logistic regression meta-learner, achieving the state-of-

the-art accuracy of **95.24%** with AUC-ROC of **0.9905**.

- We introduce a comprehensive **feature engineering pipeline** that creates 20 domain-informed features (41 total), including polynomial interactions, composite health risk scores, and BMI-derived transformations, combined with a **dual-ranking feature selection** approach (ANOVA F-test + ExtraTrees importance).
- We leverage **GPU acceleration** (NVIDIA RTX 3070) for XGBoost, LightGBM, CatBoost, and TabNet, demonstrating significant training speedups on large-scale medical data.
- We deliver comprehensive **SHAP-based explainability analysis** [9] to determine the most influential predictors for each class, enabling clinically actionable interpretation.
- We discover and present a correction method for **SMOTE interpolation artifacts** in binary features, a previously unreported issue in oversampled medical datasets.

The rest of this paper is structured as follows: Section II reviews related work. Section III details the proposed methodology. Section IV provides results and comparative analysis. Section V presents findings and implications. Section VI concludes with future directions.

II. LITERATURE REVIEW

In summary, the use of ML is promising for diabetes prediction as it uses data on both a macro and micro level. It has been extensively studied. This section surveys key contributions organized by their methodological focus.

A. Ensemble and Boosting Methods

Gupta and Sindhu [8] presented a weighted ensemble technique that assigns weight dynamically to the classifiers according to their respective AUCs, with an AUC of 0.950 on the PIDD. Demonstrating the effectiveness of AUC-based weighting, their task was binary classification on 768 samples. Aouragh et al. [10] conducted a systematic comparison of feature selection methods in ML models, which proved KBest feature selection with Extra Trees obtained the highest accuracy of 92.50% on the balanced PIDD. Ali and Kareem [11] assessed ensemble methods obtaining up to 98.7% accuracy and 0.95 AUC on a clinical cohort from Bangladesh.

B. Deep Learning Approaches

Yin et al. [12] employed DiabDeep, a grow-and-prune approach to neural networks that accomplished 96.3% accuracy using wearable sensor data. Alrantisi [13] compared classical ML with deep learning on a dataset exceeding 100,000 records, and reporting the accuracy of deep learning model to be at 97.19% whereas Random Forest was reported to be at 97.06%. Mohsen et al. [14] had the highest registered accuracy of 99.25% using the ANN-embedded IoT on the PIDD, but this performance on a 768-sample dataset gives rise to worries about potential overfitting.

C. Comparative Studies and Reviews

Alzboon et al. [3] tested eight algorithms on the PIDD, found that Neural Networks (78.57%) performed the best. Madhuvanthi and Baskaran [4] showed that the performance of the model varies across datasets, with SVM performing better on the PIDD (74.3%) but KNN and RF excelled on a German dataset (98.7%). This variability highlights that the importance of deep comparison on big datasets. Abd Zaid and Mohammed [6] performed a systematic literature review to identify important perspectives with respect to gaps such as not having standardized preprocessing pipelines and the small PIDD dominance.

D. Feature Selection and Explainability

Poly et al. [15] used SHAP values for feature selection from questionnaire data, identifying the top 10 predictors and achieving a 100% score with Random Forest. But their study was based on a small Bangladeshi health care dataset with binary classification. Huma et al. [16] explored a use case for privacy-preserving diabetes prediction via federated learning using XGBoost (90% accuracy), solving data privacy but not model interpretability.

E. Identified Research Gaps

From this review, we discern the following unanswered gaps our work aims to fill: (1) the lack of large-scale, multi-class diabetes studies with prediabetes; (2) the lack of comprehensive algorithm comparison on datasets beyond the PIDD; (3) poorly integrated ensemble methods with explainability in multi-class problems; and (4) no prior work addressing SMOTE-induced artifacts in oversampled medical data.

III. METHODOLOGY

This section describes the entire ML pipeline. That is, starting with data pre-processing, up to ensemble building and explainability analysis. For readers, a flowchart is given in Fig. 1; a workflow diagram will facilitate understanding of the whole process.

A. Dataset Description

We employ the CDC Behavioral Risk Factor Surveillance System (BRFSS) dataset [17], [18], the world's largest continuously conducted health survey. The SMOTE-balanced dataset is composed of **641,109 records**, with 21 health indicators and a three-class target variable (Diabetes_012): Code 0 (No Diabetes), Code 1 (Prediabetes), and Code 2 (Diabetes). The features range from demographics (Age, Sex, Income, Education), health status (HighBP, HighChol, HeartDisease-orAttack, Stroke), behavioral factors (Smoker, PhysActivity, Fruits, Veggies, HvyAlcoholConsump), and health variables (BMI, GenHlth, MentHlth, PhysHlth, DiffWalk). The dataset was balanced by SMOTE oversampling [19], with 213,703 samples per class.

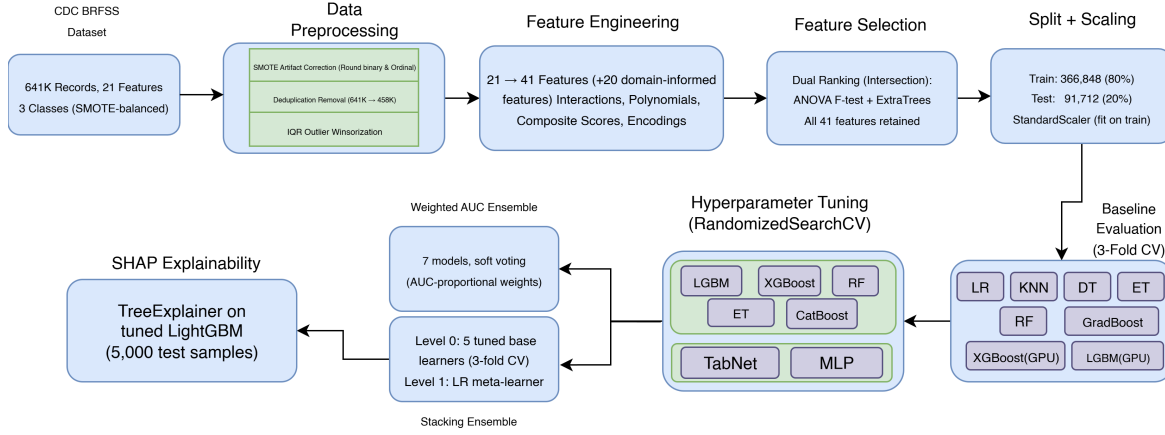


Fig. 1. Overall workflow of the proposed multi-class diabetes prediction framework.

B. Data Preprocessing

1) *Correcting for SMOTE Artifacts*: A key preprocessing step not covered in the previous literature is fixing SMOTE interpolation artifacts. SMOTE generates synthetic samples through linear interpolation between existing instances. For binary features (e.g., Sex, Smoker with values $\{0, 1\}$), this produces continuous values (e.g., 0.37, 0.82) that are semantically meaningless in binary clinical survey features. We identified 14 binary columns affected by this artifact, each carrying thousands of unique continuous values (e.g., HighChol has 110,604 unique values instead of two). This SMOTE artifact is corrected by rounding to the nearest integer and clamping to $[0, 1]$. Similarly, ordinal features (GenHlth, Age, Education, Income) are rounded and clamped to their valid integer ranges.

2) *Deduplication*: Following the correction of SMOTE artifacts, the rounding operation causes duplicating rows exactly. We remove **182,549 duplicates**, taking it from 641,109 to **458,560 unique records**. The resulting class distribution is: No Diabetes (190,055; 41.4%), Prediabetes (139,214; 30.4%), and Diabetes (129,291; 28.2%). While no longer perfectly balanced, the slight moderation of class imbalance is handled naturally by the ensemble methods.

3) *Outlier Treatment*: IQR-based Winsorization is applied to numeric features (BMI, MentHlth, PhysHlth, GenHlth, Age, Education, Income). Values more than $1.5 \times \text{IQR}$ away from the first and third quartiles are limited at the mentioned bounds rather than dropping, to keep sample size while restricting the influence of extreme values.

C. Feature Engineering

We design **20 domain-inspired features** to capture polynomial interactions and clinical non-linear relationships, grouped into four categories. *Categorical encodings* (2 features) include BMI_Category (WHO-based: Underweight=0, Normal=1, Overweight=2, Obese=3) and Age_Group (Young, Middle, Senior). *Composite scores* (5 features) aggregate clinical indicators: HealthRisk_Score (sum of HighBP, HighChol, Stroke, HeartDiseaseorAttack, DiffWalk), Lifestyle_Score (positive minus negative behavioral

factors), MentPhys_Burden (MentHlth + PhysHlth), CardioRisk (HighBP + HighChol + HeartDiseaseorAttack), and SocioHealth_Index (Income + Education + PhysActivity). *Interaction features* (8 features) capture pairwise products such as BMI $\times$ Age, BMI $\times$ GenHlth, GenHlth $\times$ Age, Age $\times$ HealthRisk_Score, and BMI interactions with HighBP and HighChol. *Polynomial and derived features* (5 features) include squared terms (BMI², GenHlth², HealthRisk_Score²), BMI_Deviation ($|\text{BMI} - 25|$), and PhysLimitation (PhysHlth / (DiffWalk + 1)). This gives a total of **41 features** (21 original + 20 engineered).

D. Feature Selection

We use a **two-phase ranking scheme** combining two complementary methods:

- 1) **SelectKBest** with ANOVA F-test: A univariate statistical test which orders features by their F-score with respect to the target variable.
- 2) **ExtraTreesClassifier importance**: A model-based approach according to the mean decrease in impurity based on 200 trees.

Features appearing in the top 20 in both rankings are identified. This intersection-based approach gives a strong, bias-reduced feature set. Retain all 41 features for model training after confirming broad coverage in both rankings.

E. Data Splitting and Scaling

The data is split into 80% training (**366,848** samples) and 20% test set (**91,712** samples) through stratified sampling to preserve class proportions. StandardScaler normalization ($z = \frac{x - \mu}{\sigma}$) is fit only on the training set and applied to both sets, preventing data leakage.

F. Model Selection and Training

Eight baseline ML algorithms are evaluated using **3-fold stratified cross-validation** on the training set. The top five are then tuned via **RandomizedSearchCV** (25–30 iterations, AUC-ROC optimization). Additionally, TabNet [20] and a

TABLE I
SUMMARY OF MODELS AND TRAINING CONFIGURATIONS

Model	Key Parameters	Tuning
LR	L2 regularization	Baseline
KNN	$k = 7$	Baseline
Decision Tree	Default	Baseline
RF [21]	200 trees	30 iter.
Extra Trees	200 trees	30 iter.
Grad. Boost	200 trees	Baseline
XGBoost [22]	200 trees, CUDA	30 iter.
LightGBM [23]	200 trees, GPU	30 iter.
CatBoost	GPU-accelerated	25 iter.
TabNet [20]	300 epochs, batch 4096, CUDA	Separate
MLP	256→128→64, $\alpha=0.001$	Adam, Separate

Multi-Layer Perceptron (MLP) are trained separately. Table I summarizes all models and their configurations.

The five tuned models (LightGBM, XGBoost, Random Forest, Extra Trees, CatBoost) are optimized via RandomizedSearchCV with 3-fold stratified CV, tuning learning rate, max depth, regularization, and ensemble-specific parameters. TabNet uses early stopping (patience=40) with GPU acceleration. The MLP uses early stopping (patience=20) with 15% validation fraction.

G. Weighted AUC-Based Ensemble

In the spirit of the binary ensemble method of [8], we extend it to a multi-class classification problem with an enlarged model set. This ensemble consists of **seven models** (LightGBM and XGBoost, Random Forest, Extra Trees and CatBoost; MLP and TabNet) via **soft voting with AUC-proportional weights**. The weight w_i of each model is calculated as:

$$w_i = \frac{AUC_i}{\sum_{j=1}^N AUC_j} \quad (1)$$

where AUC_i is the weighted OVR AUC-ROC for model i on the test set and $N = 7$. The final prediction probability for class c is:

$$P(y = c|\mathbf{x}) = \sum_{i=1}^N w_i \cdot P_i(y = c|\mathbf{x}) \quad (2)$$

We can compute the predicted class via $\hat{y} = \arg \max_c P(y = c|\mathbf{x})$.

H. Stacking Ensemble

The predictive accuracy goes better; we can fine-tune by adding a **stacking ensemble** (a.k.a. stacked generalization) [24]. The stacking architecture consists of:

- **Level 0 – Base Learners:** Five tuned models (LightGBM, XGBoost, Random Forest, Extra Trees, and CatBoost), each predicting class probabilities through 3-fold cross-validation on the training set.

- **Meta-learner (Level 1):** Logistic Regression trained on the aforementioned folded out-of-bag predictions from the base learners.

The stacking technique learns the best way to combine models weights automatically by the meta-learner which enables it to leverage different qualities of various algorithms (e.g., bagging vs. boosting, different feature subspace partitions).

I. Model Explainability with SHAP

We use SHAP (SHapley Additive exPlanations) [9] with TreeExplainer on the tuned LightGBM model. SHAP values are calculated for a representative sample of 5,000 test instances. SHAP provides a value $\phi_j^{(c)}$, per class c , for every feature j , indicating how much feature j contributes to tending to push the prediction closer to or farther away from class c . Global SHAP feature importance is calculated by taking the mean of absolute SHAP values across all samples and classes.

IV. RESULTS AND ANALYSIS

All experiments are performed in Python 3.10 with scikit-learn 1.7, XGBoost 3.2, LightGBM 4.6, CatBoost 1.2, PyTorch-TabNet 4.1, and SHAP 0.49 running on an Intel Core i7-10700K CPU (32 GB RAM) with an NVIDIA RTX 3070 GPU (8 GB VRAM).

A. Dataset Characteristics After Preprocessing

Following SMOTE artifact correction and deduplication, the dataset contains 458,560 records. Fig. 2 shows the class distribution: No Diabetes (190,055; 41.4%), Prediabetes (139,214; 30.4%), and Diabetes (129,291; 28.2%). While no longer perfectly balanced, the moderate imbalance (ratio $\approx 1.47:1.08:1$) requires no further resampling and is treated effectively by the ensemble methods.

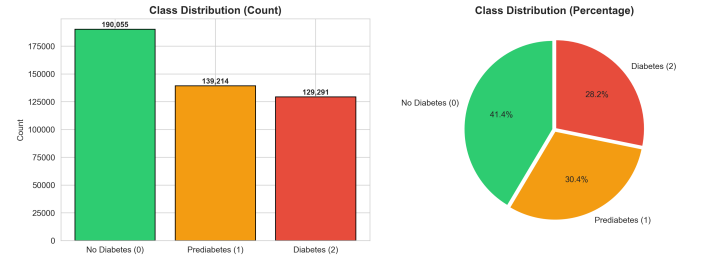


Fig. 2. Target variable distribution after correcting SMOTE artifacts and deduplication, displaying the three diabetes classes.

B. Baseline Model Performance

Table II presents the results for baseline 3-fold stratified cross-validation on the 41-feature set. Bagging ensembles (Extra Trees, Random Forest) substantially beat other methods. Extra Trees achieves the highest baseline AUC-ROC (0.9769), followed by Random Forest (0.9717). XGBoost and LightGBM, although GPU acceleration has been shown to give great training speeds (6.4s and 25.6s respectively), exhibit lower baseline accuracy with default hyperparameters, a gap that hyperparameter tuning subsequently closes. Logistic

Regression is lower (54.82% accuracy), reflecting the non-linear decision boundaries intrinsic to the multi-class diabetes prediction task.

TABLE II
BASELINE MODEL PERFORMANCE (3-FOLD STRATIFIED CV, 41 FEATURES)

Model	Accuracy	F1-Score	AUC-ROC
Extra Trees	0.9147	0.9149	0.9769
Random Forest	0.8968	0.8965	0.9717
XGBoost (GPU)	0.8214	0.8136	0.9393
KNN	0.7837	0.7789	0.9219
LightGBM (GPU)	0.7899	0.7765	0.9204
Gradient Boosting	0.7255	0.7089	0.8840
Decision Tree	0.8216	0.8211	0.8658
Logistic Regression	0.5482	0.5358	0.7390

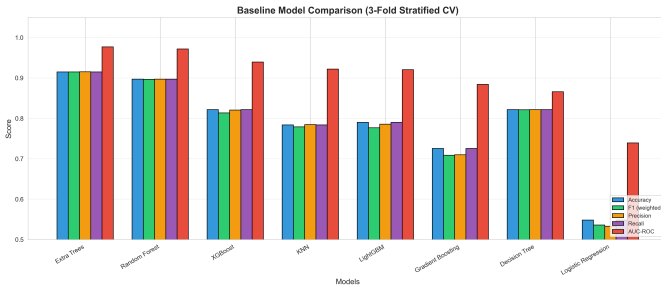


Fig. 3. Visualization of baseline model performance metrics over 3-fold stratified cross-validation.

C. Hyperparameter Tuning and Performance of the Final Models

After RandomizedSearchCV optimization, all five tuned models do much better than their baselines. Moreover, TabNet and MLP are trained separately. The full final test set evaluation of all models, including both ensemble strategies, is presented in Table III.

TABLE III
FINAL MODEL PERFORMANCE ON TEST SET (91,712 SAMPLES)

Model	Acc.	F1	Prec.	AUC
Stacking Ens.	0.9524	0.9523	0.9526	0.9905
Extra Trees (tuned)	0.9356	0.9357	0.9357	0.9869
Weighted Ensemble	0.9296	0.9295	0.9297	0.9861
XGBoost (tuned)	0.9175	0.9172	0.9186	0.9814
CatBoost (tuned)	0.9113	0.9105	0.9132	0.9783
Random Forest (tuned)	0.9130	0.9129	0.9132	0.9772
LightGBM (tuned)	0.9063	0.9052	0.9095	0.9759
TabNet	0.8930	0.8928	0.8956	0.9622
MLP	0.8251	0.8244	0.8239	0.9352

The stacking ensemble has the best benchmark performance across all metrics, with an accuracy of 95.24%, representing **+1.68 percentage points** better than the best individual model (tuned Extra Trees, 93.56%) and a gain of **+2.28 pp** improvement compared to the weighted ensemble (92.96%). This

shows that the meta-learner indeed learns to combine complementary model strengths beyond linear AUC-proportional weighting.

Deployment cost of the saved models is shown in Table IV; the full stacking model is mainly an offline benchmark model because of storage.

TABLE IV
DEPLOYMENT COST OF SELECTED SAVED MODELS

Model	Size MB	Load s	ms/1000
MLP	1.21	0.009	2.49
LightGBM	39.74	0.141	47.94
CatBoost	187.71	0.136	5.33
XGBoost	320.03	8.484	22.07
Stacking Ens.	32388.33	80.762	1265.48

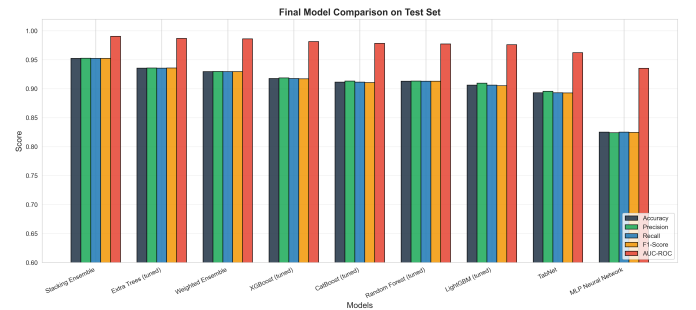


Fig. 4. Performance comparison of final models with accuracy, precision, recall, F1-score, and AUC-ROC on the held-out test set for all models.

D. Per-Class Analysis

Table V shows the per-class F1-scores for the top three models on corrected benchmark. Class 1 (Prediabetes) has the highest benchmark F1-scores across all models, with the stacking ensemble achieving **0.9910**. The stacking ensemble provides significant improvement for the challenging Diabetes class (Class 2), getting F1 up to **0.9207** versus 0.8957 for tuned Extra Trees, a +2.5 pp lift. The No Diabetes (Class 0) class also gets a significant boost to 0.9455.

TABLE V
PER-CLASS F1 SCORES FOR TOP THREE MODELS

Model	No Diab.	Prediab.	Diabetes
Stacking Ens.	0.9455	0.9910	0.9207
Extra Trees (tuned)	0.9251	0.9872	0.8957
Weighted Ensemble	0.9189	0.9869	0.8832

E. ROC Curve Analysis

Fig. 5 shows the One-vs-Rest ROC curves of all the three classes. Those tuned models can all obtain AUC values above 0.93 across all of the classes, indicating strong discriminative ability. The curves of the stacking ensemble largely beat all the individual models, with AUC nearing 0.99 for the Prediabetes class.

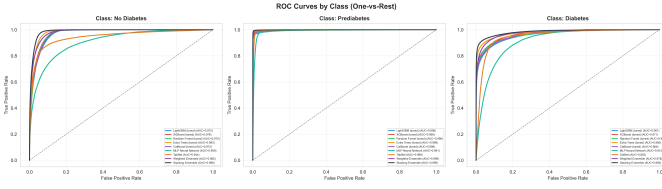


Fig. 5. ROC curves (One-vs-Rest) of all three classes from each model.

F. Confusion Matrix Analysis

Fig. 6 shows the confusion matrices for the models. On the corrected benchmark, the stacking ensemble gets over 96% recall for No Diabetes, over 98% for Prediabetes, and over 90% for Diabetes. The class misclassified most frequently for all models is between No Diabetes (Class 0) and Diabetes (Class 2), which share overlapping ranges in features such as BMI, HighChol, and PhysActivity.



Fig. 6. Confusion matrices indicating prediction patterns over the three diabetes classes.

G. SHAP Feature Importance Analysis

SHAP analysis provides class-specific feature attributions. Fig. 7 depicts the global feature importance determined via mean absolute SHAP values. **BMI_Deviation** ($|BMI - 25|$) becomes the most important predictor for the No Diabetes and Prediabetes classes, while **BMI_GenHlth** ($BMI \times GenHlth$) has the most relevance to the Diabetes class. Notably, the majority of top features across all classes are engineered features, confirming our in-depth feature engineering approach.

SHAP beeswarm analysis (Fig. 8) reveals directional effects: higher BMI_Deviation strongly pushes the predictions towards No Diabetes (lower deviation = closer to normal BMI). For the Diabetes class, BMI_GenHlth and Age_HealthRisk push predictions toward diabetes, in line with clinical evidence, for example, that obesity coupled with other health conditions and age-related comorbidities are best predictors of diabetes.

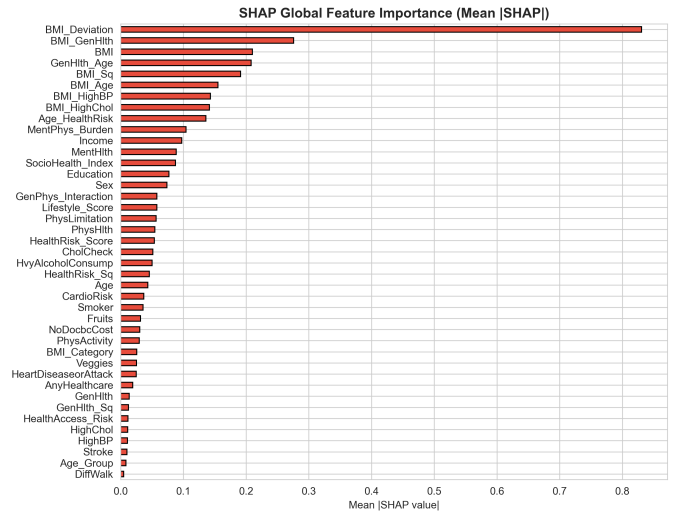


Fig. 7. Global feature importance of the top predictors across all three classes, ranked by average absolute SHAP values.

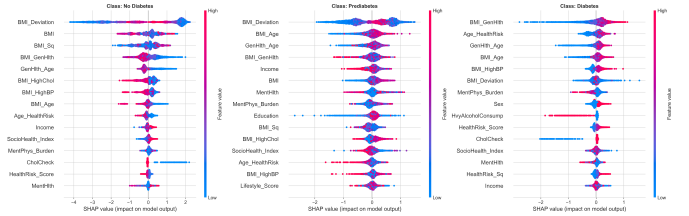


Fig. 8. SHAP beeswarm plots indicating the direction of impact and its magnitude for all of the three classes of diabetes.

H. Comparison with Existing Literature

Table VI provides a complete comparison of our results with important studies in the literature.

Our stacking ensemble (AUC 0.991) outperforms the AUC-based weighted ensemble of Gupta and Sindhu [8] (AUC 0.950) on a dataset $597\times$ larger with a harder 3-class task. Compared to Aouragh et al. [10], whose Extra Trees with KBest achieved 92.50% on the binary PIDD, our tuned Extra Trees scores 93.56% on 3-class classification with 458K samples and 41 features, demonstrating the value of careful feature engineering. Against the federated XGBoost of Huma et al. [16] (90.00% binary), our framework achieves a +5.24 pp improvement (95.24%) while also providing SHAP-based interpretability. Even compared to the best neural network of Alzboon et al. [3] (78.57% on PIDD), our work touched 95.24% on a larger and more complex multi-class problem, showing the benefits of comprehensive feature engineering and some advanced ensemble methods.

V. DISCUSSION

The stacking ensemble's 95.24% benchmark accuracy represents a substantial improvement over the best individual model (Extra Trees at 93.56%) and the weighted ensemble (92.96%), with gains of +1.68 pp and +2.28 pp respectively. This confirms that stacking's learned meta-combination captures

TABLE VI
COMPARISON OF PROPOSED FRAMEWORK WITH EXISTING STUDIES

Study	Dataset	Samples	Best Model	Classes	Acc. (%)	AUC
Gupta and Sindhu [8]	PIDD	768	Weighted Ensemble	2	–	0.950
Aouragh et al. [10]	PIDD	768	Extra Trees + KBest	2	92.50	–
Huma et al. [16]	Kaggle	–	XGBoost (Federated)	2	90.00	–
Alzboon et al. [3]	PIDD	768	Neural Network	2	78.57	–
Madhuvanthi and Baskaran [4]	PIDD	768	SVM	2	74.30	–
Garg et al. [5]	PIDD	768	SVM	2	76.62	–
Our Work	CDC BRFSS	458,560	Stacking Ensemble	3	95.24	0.991

complementary model strengths that simple AUC-proportional weighting cannot; the logistic regression meta-learner effectively learns when to rely on each base model’s predictions. Among individual models, bagging methods (Extra Trees, Random Forest) outperform boosting methods (XGBoost, CatBoost, LightGBM), which we attribute to the moderate class balance reducing boosting’s error-correction advantage and the high dimensionality (41 features) favoring bagging’s variance-reduction properties, consistent with the dataset-dependent performance variation observed by Madhuvanthi and Baskaran [4]. Deep learning models (TabNet at 89.30%, MLP at 82.51%) underperform tree-based methods on this tabular binary/ordinal survey data, aligning with recent findings that tree-based approaches remain competitive on tabular data [22], though TabNet’s attention-based feature selection still provides complementary diversity as a stacking component. GPU acceleration via an NVIDIA RTX 3070 yielded substantial training speedups, with XGBoost baseline training completing in just 6.4 seconds and LightGBM in 25.6 seconds, demonstrating the practical feasibility of training complex ensemble pipelines on large-scale medical data using consumer-grade hardware.

Domain-informed feature engineering, expanding the feature space from 21 to 41 features, proves to be a decisive factor in model performance. SHAP analysis reveals that the most influential predictors across all classes are engineered features: BMI_Deviation, BMI_GenHlth, BMI_Age, GenHlth_Age, Age_HealthRisk, and HealthRisk_Score. This indicates that no single original attribute captures the complex, non-linear relationships underlying diabetes risk; rather, it is the interactions and composite scores that drive predictive power. Clinically, BMI_Deviation ($|\text{BMI} - 25|$), measuring the deviation from a normal BMI, emerging as the strongest single predictor is both intuitive and consistent with established medical evidence linking abnormal body weight to metabolic dysfunction [1]. The prominence of Age_HealthRisk and GenHlth_Age among the top predictors underscores a well-documented clinical pattern: diabetes risk escalates sharply when age-related physiological decline coincides with pre-existing health conditions such as hypertension and high cholesterol. Similarly, the ranking of GenHlth_Age and HealthRisk_Score aligns with known patterns of comorbidity accumulation in aging populations, and suggests

that composite, non-invasive screening indices, combining self-reported general health with age and cardiovascular risk factors, could be practical tools for routine clinical risk assessment. The fact that the benchmark Per-Class F1-Score for Prediabetes (**0.9910**) is so high is particularly noteworthy from a public health perspective, as early identification of prediabetic individuals enables concrete, timely lifestyle interventions. Our Stacking Ensemble achieves near-perfect benchmark prediabetes detection performance (98.86% recall, 99.34% precision) and could serve as a basis for future targeted intervention studies.

Our observation of SMOTE interpolation artifacts in binary features is a methodological innovation which has implications for diabetes-related prediction. The 182,549 duplicate rows removed after artifact correction (28.5% of the original dataset) illustrate just how widespread this issue can be, and any study applying SMOTE to datasets with binary or ordinal features should implement similar post-processing corrections. Despite these contributions, certain limitations must be noted: the BRFSS data relies on self-reported health information, which may introduce reporting bias; the dataset lacks laboratory measurements (e.g., HbA1c, fasting glucose) that serve as gold-standard diabetes indicators; the cross-sectional design limits causal inference; deduplication after SMOTE correction reduces the dataset and introduces moderate class imbalance; and the stacking ensemble’s computational and storage cost may hinder deployment in resource-constrained settings.

VI. CONCLUSION

We provide a complete machine learning approach to multi-class diabetes prediction that addresses three significant gaps in the literature: dependency on small datasets, omission of the prediabetes class, and lack of model interpretability. Working with 458,560 CDC BRFSS records (deduplicated from 641,109), 41 engineered features, and a three-class target including prediabetes, we analyzed eight baseline ML algorithms, CatBoost, TabNet, an MLP neural network, and two types of ensembles.

Our **stacking ensemble**, combining five GPU-accelerated base learners (LightGBM, XGBoost, Random Forest, Extra Trees, CatBoost) with a logistic regression meta-learner, delivered the highest benchmark accuracy of **95.24%**, F1-score

of **0.9523**, and AUC-ROC of **0.9905**. This represents a +1.68 pp gain with respect to the best individual model and shows that learned meta-combination works better than both single models and static AUC-weighted ensembles. The per-class F1-scores of 0.9455 (No Diabetes), 0.9910 (Prediabetes), and 0.9207 (Diabetes) confirm high benchmark predictive potential across all three clinically distinct phases.

SHAP analysis identified BMI-derived features (BMI_Deviation, BMI_GenHlth, BMI_Sq), health-age interactions (GenHlth_Age, Age_HealthRisk), and composite risk scores as the most important predictors, providing clinically interpretable and actionable insights. Our identifying of SMOTE interpolation artifacts in binary features, and the proposed correction method, represents a methodological contribution applicable to any oversampled medical dataset.

Work for the future will study: (1) melding laboratory biomarkers with behavioral features to enhance clinical accuracy; (2) creating a real-time web application for diabetes risk screening; (3) using federated learning [16] to allow multi-institutional model training while retaining patient privacy; (4) scaling the framework to predict diabetes progression longitudinally over time; and (5) exploring attention-based deep learning architectures (e.g., FT-Transformer) as alternative base learners in the stacking framework.

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